

# Transcorrelated Jastrow Fitting: Atomization Energies

## 1 Theory

The procedure described below is motivated by the reference energy correction scheme in TC theory [cite johannes], that states that the basis set error in correlation energy converges quickly with TC, while the errors in the reference energy (i.e., the Hartree-Fock energy) is untouched. To avoid calculations in multiple basis sets and make the workflow more simple and practical, we solve the reference state in a large basis set, and then construct a virtual space with size equal to a lower-level basis. This keeps the reference accurate, while allowing us to perform the correlation calculation in a smaller virtual space that is more similar to the small basis set.

The selected virtual space is constructed by projecting the large-basis virtual MOs onto the small-basis AO space via

$$M = S_{11}^{-1/2} S_{12} C^{\text{virt}} \quad [n_{\text{small}} \times n_{\text{large,virt}}] \quad (1)$$

where  $S_{11}$  and  $S_{12}$  are the small-basis and cross-overlap matrices, and  $C^{\text{virt}}$  collects the large-basis virtual MO coefficients. The elements  $M_{\mu a} = \langle \tilde{\varphi}_\mu | \psi_a \rangle$  are overlaps between Löwdin-orthonormalized small AOs and  $S_{22}$ -orthonormal large-basis virtuals, so the thin SVD

$$M = U \Sigma V^\top \quad U [n_{\text{small}} \times n_{\text{small}}], \Sigma [n_{\text{small}} \times n_{\text{small}}], V [n_{\text{large,virt}} \times n_{\text{small}}] \quad (2)$$

yields singular values  $\sigma_k \leq 1$  that are exact cosines of the principal angles between the two subspaces. Dropping the  $n_{\text{occ}}$  columns of  $V$  with near-zero  $\sigma_k$  and projecting onto the AO basis gives the  $n_{\text{sel}} = n_{\text{small}} - n_{\text{occ}}$  selected virtuals

$$C^{\text{sel}} = C^{\text{virt}} V_{\text{kept}} \quad [n_{\text{large,AO}} \times n_{\text{sel}}] \quad (3)$$

which are  $S_{22}$ -orthonormal by construction. The SVD rotation mixes canonical virtuals, rendering the projected Fock matrix

$$F^{\text{sel}} = V_{\text{kept}}^\top \text{diag}(\varepsilon_a) V_{\text{kept}} \quad [n_{\text{sel}} \times n_{\text{sel}}] \quad (4)$$

non-diagonal. Diagonalizing  $F^{\text{sel}} = W \tilde{\Lambda} W^\top$  yields the semicanonical orbitals

$$C^{\text{final}} = C^{\text{virt}} V_{\text{kept}} W \quad [n_{\text{large,AO}} \times n_{\text{sel}}]. \quad (5)$$

## 2 Atomization energies

We compute atomization energies with several xTC method families and benchmark them against SHCI+PBE+CV reference values. All calculations employ effective core potentials (ECPs) for the third-row elements. The SHCI, SHCI+PBE, and conventional CCSD(T) results used here as references are taken from Ref. [2]; the SHCI+PBE energies extrapolated to the complete basis set (CBS) limit are considered the best available benchmark for the G2/55 set. Per-basis, per-molecule method discrepancies and the full basis set convergence of error metrics are hosted as interactive plotly figures [1].

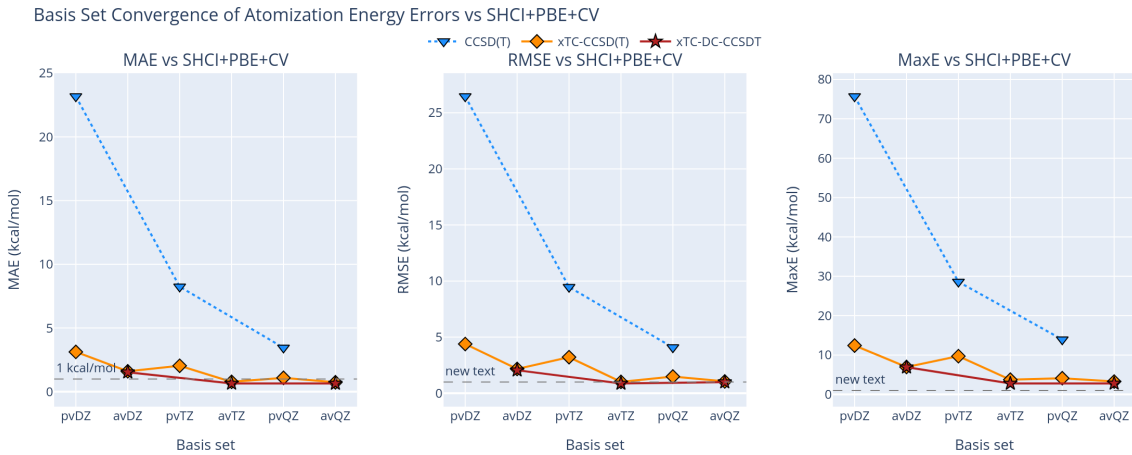


Figure 1: Basis set convergence (MAE, RMSE, MaxE in kcal/mol) of atomization energy errors vs SHCI+PBE+CV for each method family, on a unified x-axis interleaving cc-pV $x$ Z and aug-cc-pV $x$ Z. The interactive version and per-basis per-molecule breakdowns are available online [1].

Table 1: MAE, RMSE, and MaxE of atomization energies vs SHCI+PBE+CV reference (kcal/mol), for each method and basis set.

| Method              | Basis | MAE   | RMSE  | MaxE  |
|---------------------|-------|-------|-------|-------|
| (kcal/mol)          |       |       |       |       |
| <i>SHCI</i>         |       |       |       |       |
|                     | CBS   | 0.18  | 0.26  | 0.75  |
| <i>xTC-CCSD(T)</i>  |       |       |       |       |
|                     | pvDZ  | 3.13  | 4.38  | 12.40 |
|                     | avDZ  | 1.60  | 2.16  | 6.95  |
|                     | pvTZ  | 2.04  | 3.21  | 9.70  |
|                     | avTZ  | 0.78  | 1.01  | 3.73  |
|                     | pvQZ  | 1.09  | 1.48  | 4.10  |
|                     | avQZ  | 0.74  | 1.05  | 3.26  |
| <i>xTC-DC-CCSDT</i> |       |       |       |       |
|                     | avDZ  | 1.53  | 2.05  | 6.90  |
|                     | avTZ  | 0.65  | 0.87  | 2.77  |
|                     | avQZ  | 0.65  | 0.99  | 2.78  |
| <i>CCSD(T)</i>      |       |       |       |       |
|                     | pvDZ  | 23.18 | 26.47 | 75.71 |
|                     | pvTZ  | 8.25  | 9.46  | 28.60 |
|                     | pvQZ  | 3.45  | 4.08  | 13.94 |
|                     | pv5Z  | 1.67  | 1.96  | 5.52  |

## References

- [1] Kristoffer Simula. Interactive plotly figures: xtc atomization energies vs shci+pbe+cv. Online collection, GitHub Pages, 2026. <https://simulak1.github.io/tc-g2-data/>.
- [2] Yuan Yao, Emmanuel Giner, Junhao Li, Julien Toulouse, and C. J. Umrigar. Almost exact energies for the Gaussian-2 set with the semistochastic heat-bath configuration interaction method. *J. Chem. Phys.*, 153(12):124117, 2020. doi:10.1063/5.0018577.